

Inventory Management System

Student: Shashikant Chavan (24MCS10029)

Professor: Dr.Richa Sharam

University: Chandigarh Univeristy

Project Overview

- The **Inventory Management System** is a web-based application built using **Python and Django**. It provides a structured platform for managing inventory, tracking orders, and analyzing product sales. The system differentiates between two user roles: **Admin** and **Staff**, each with distinct permissions and functionalities.

Technologies used in project:

- **Backend:** Python, Django  python™ **django**
- **Frontend:** HTML, CSS  
- **Database:** MySQL (using Django ORM)  

User Roles and Functionalities

1. Admin

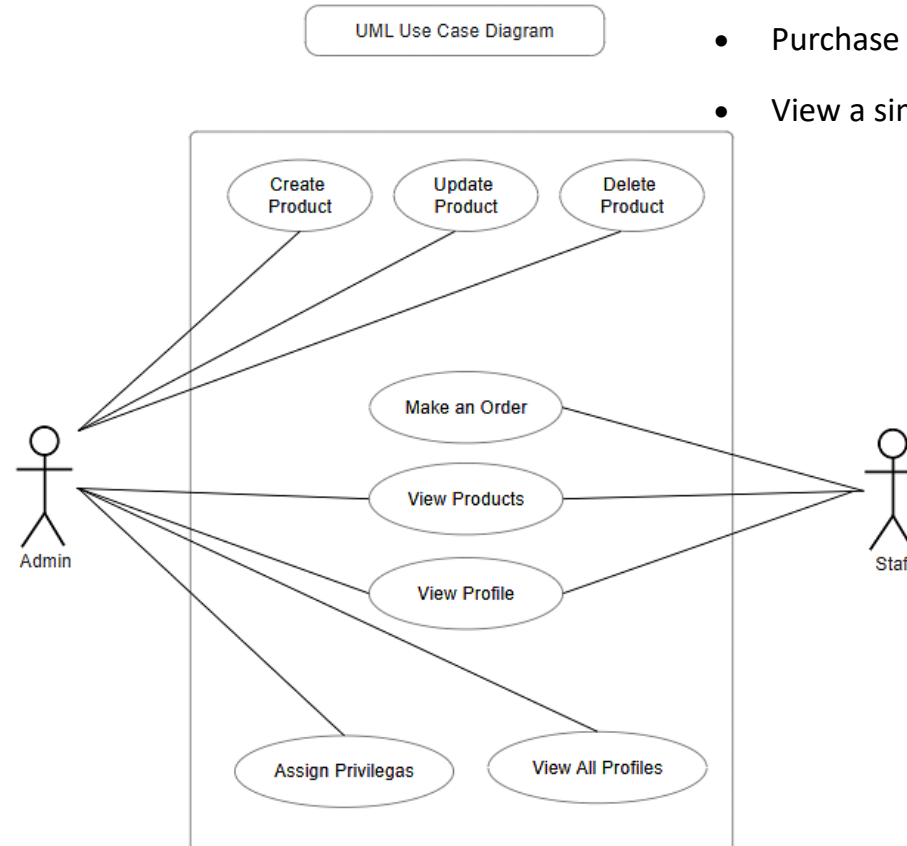
Admins have full access to the system and can perform the following tasks:

- View, edit, and delete products.
- View staff profiles.
- View and manage orders.
- Access summary charts for:
 - Products per quantity.
 - Products per order.
 - Least sold products.
 - Most sold products.

2. Staff

Staff members have restricted access and can only:

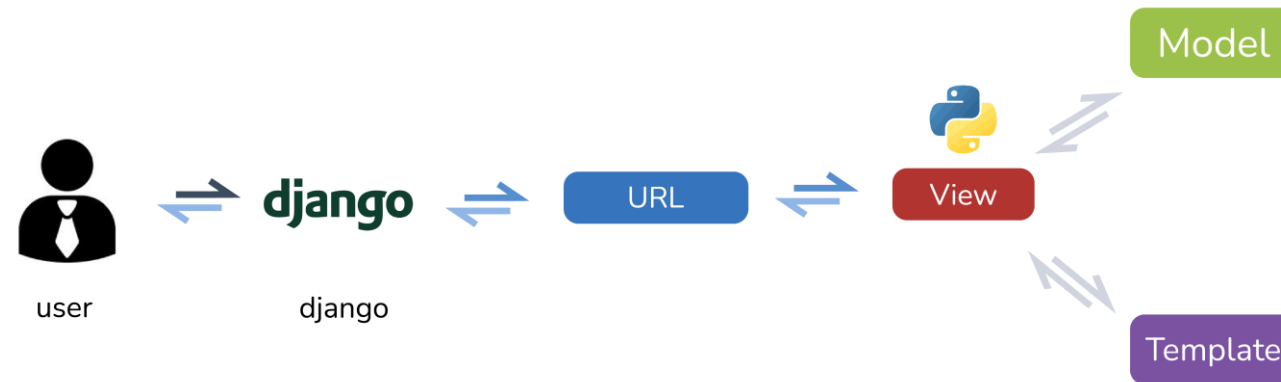
- Purchase products.
- View a simplified dashboard with relevant functionalities.



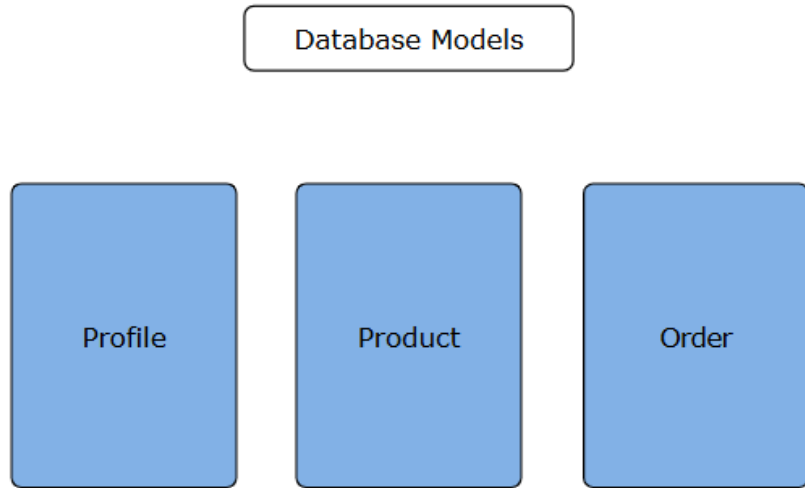
System Architecture

The system follows Django's **MTV (Model-Template-View)** architecture:

1. **Models:** Represent the database structure and define data relationships.
2. **Templates:** HTML files that handle the front-end display.
3. **Views:** Handle business logic and process user requests.



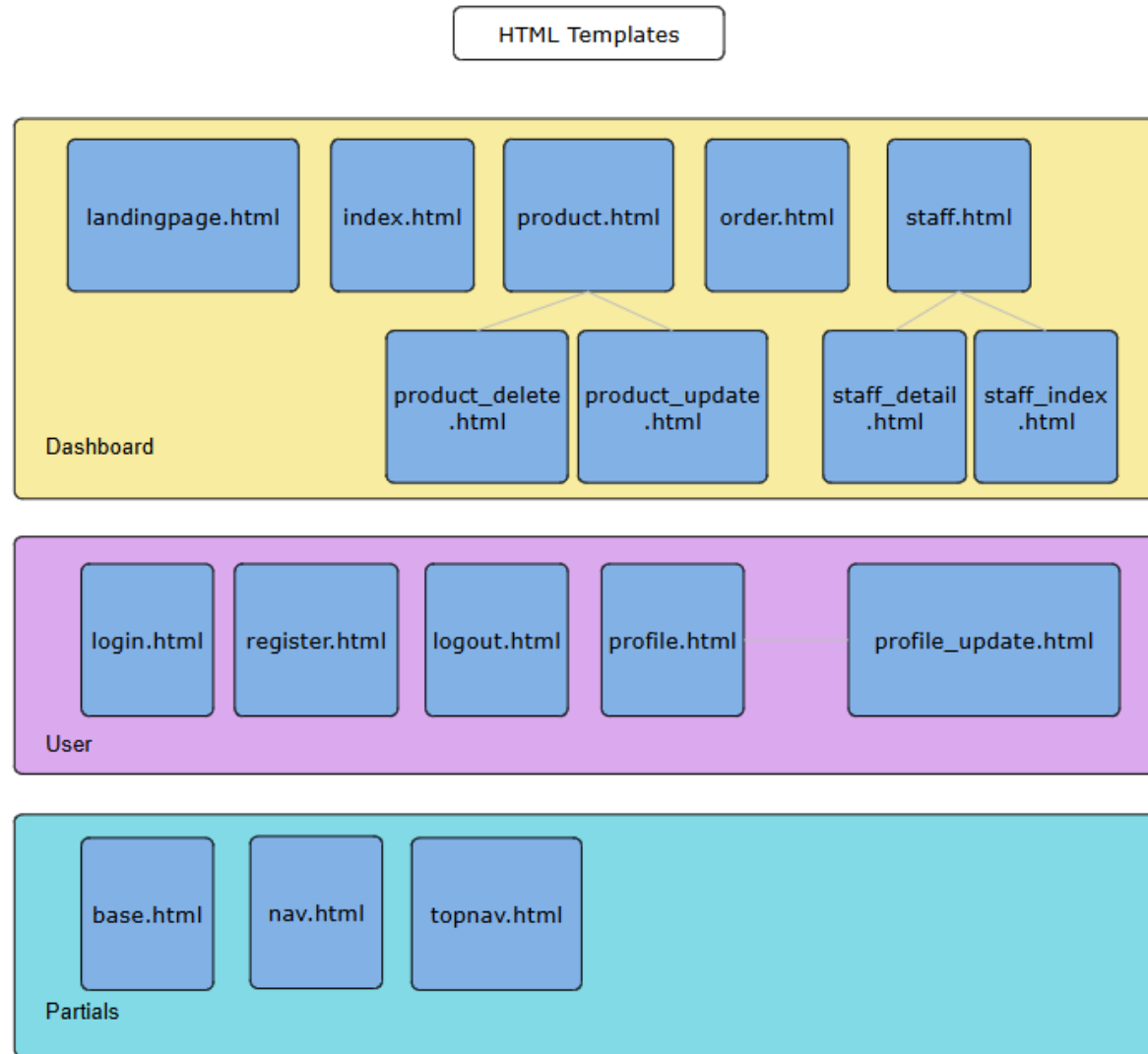
Database Models



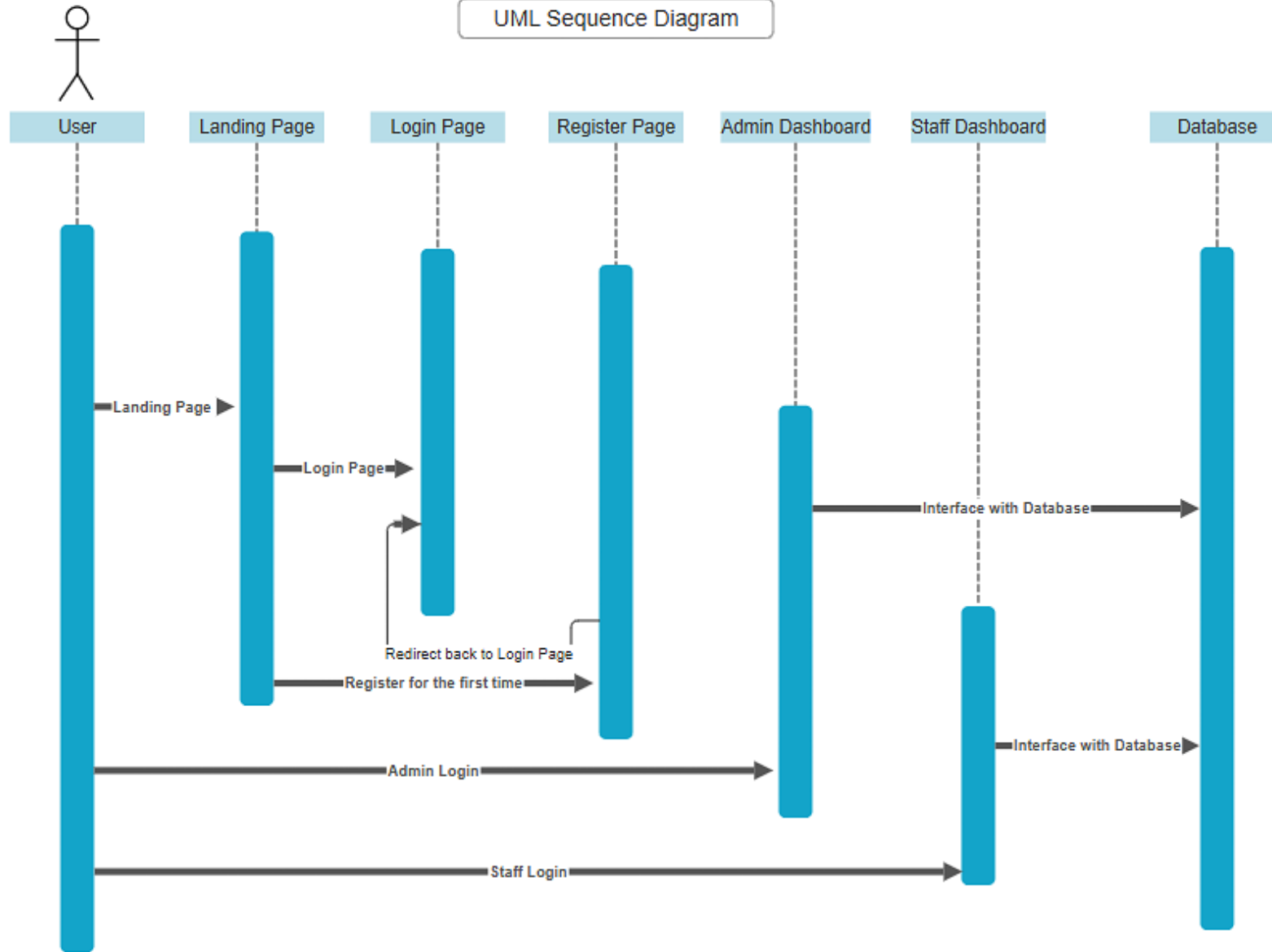
The system includes three primary models:

- **Profile:** Stores user details and role information (admin or staff).
- **Product:** Contains product information such as name, price, and stock quantity.
- **Order:** Tracks purchase details, including customer, product, quantity, and status.

Templates



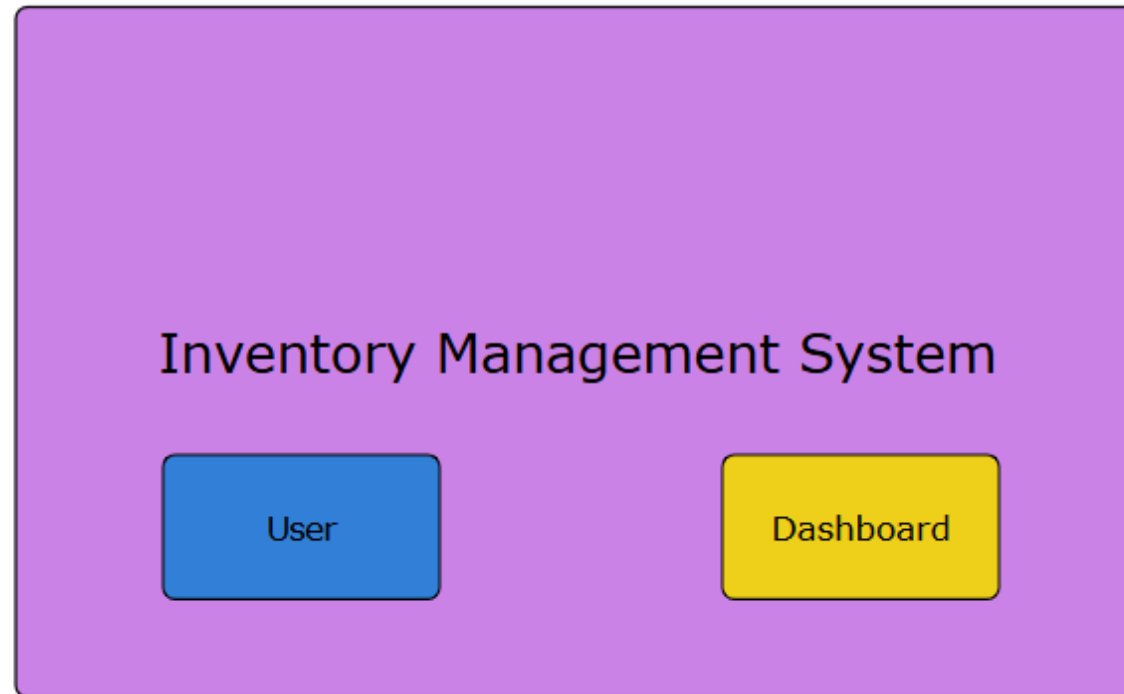
UML Sequence Diagram



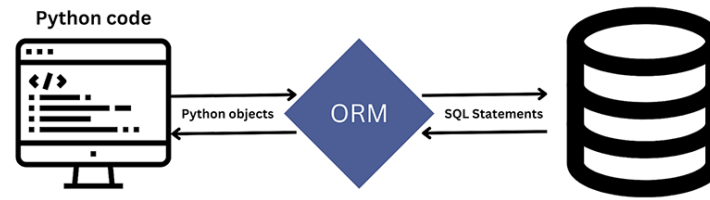
Django Apps

The system is divided into two Django apps for better modularity:

- **User:** Handles authentication, user profiles, and role management.
- **Dashboard:** Manages product inventory, orders, and analytics.

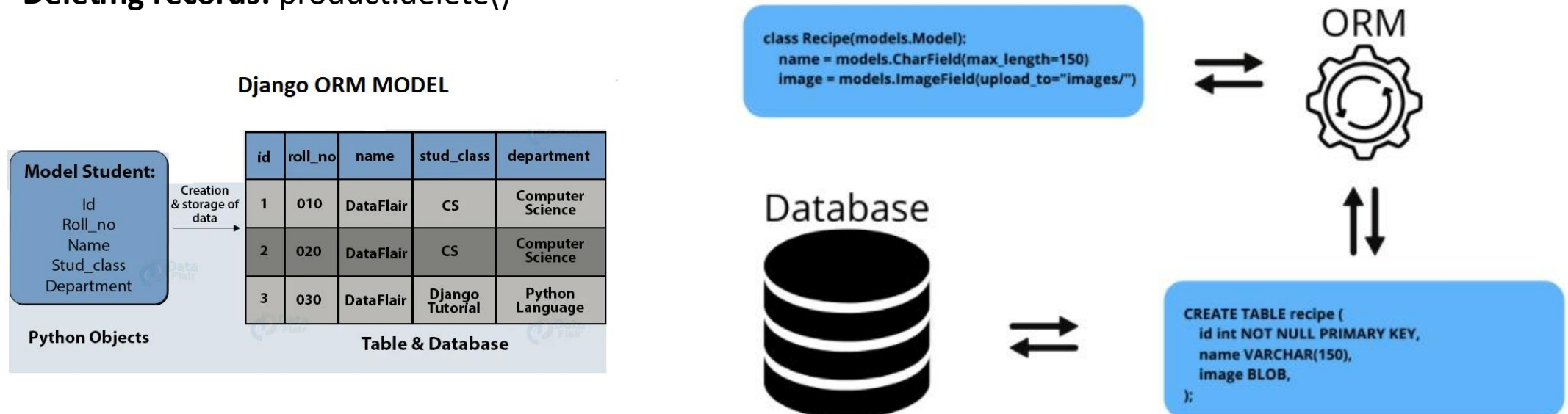


Django ORM



The project utilizes Django's **Object-Relational Mapping (ORM)** to interact with the MySQL database efficiently. Key ORM operations used:

- **Querying data:** `Product.objects.all()`, `Order.objects.filter(status='Completed')`
- **Creating records:** `Product.objects.create(name='Laptop', price=1000, stock=10)`
- **Updating records:** `product.stock -= 1; product.save()`
- **Deleting records:** `product.delete()`



Summary

This **Inventory Management System** provides an intuitive interface for admins and staff to manage inventory efficiently. By leveraging Django's powerful architecture and ORM, the system ensures smooth data management and scalability.

Screenshots

